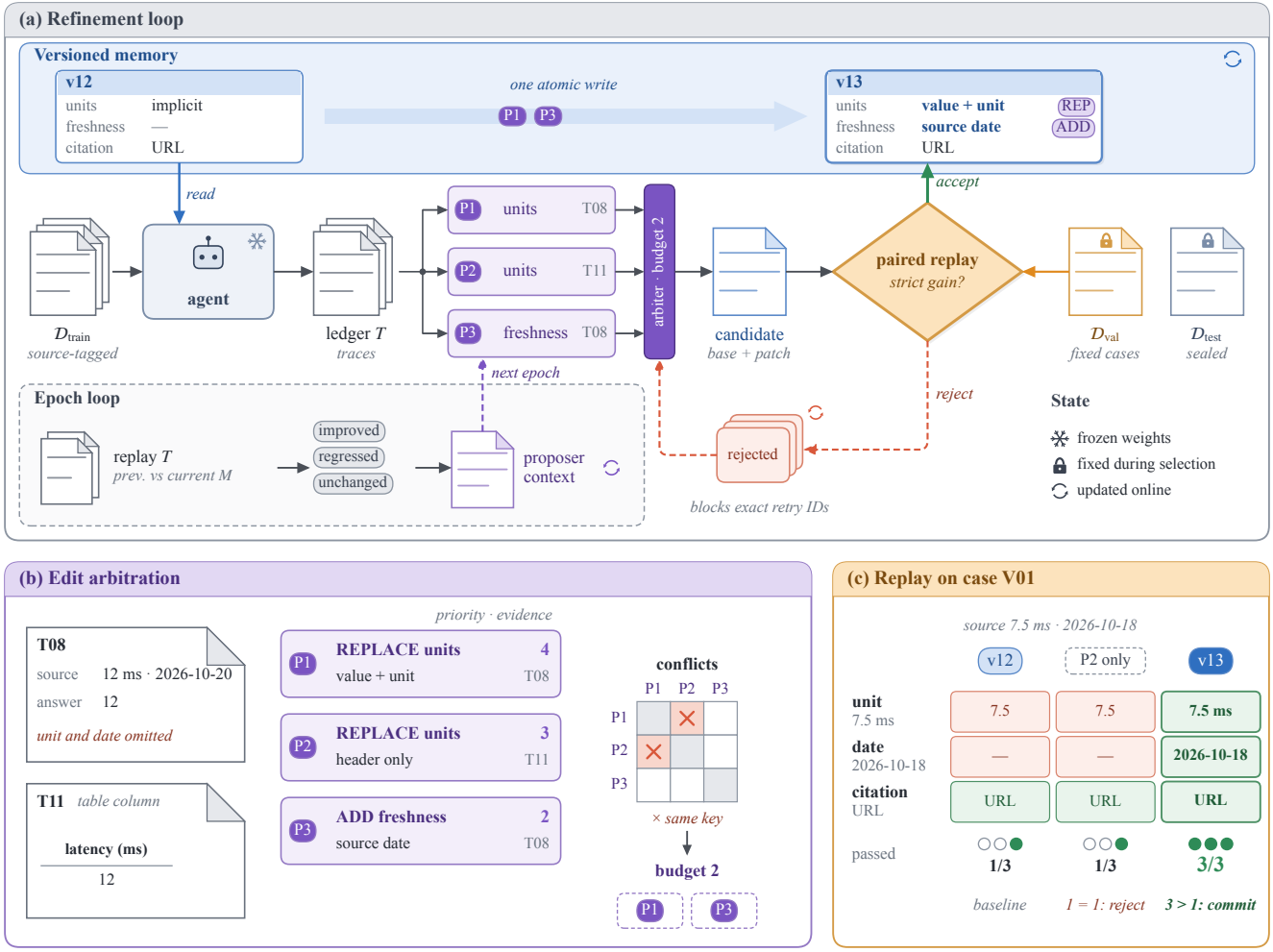

(a) Refinement loop
Versioned memory
v12
units implicit
freshness —
citation URL
one atomic write
P1 P3
v13
units value + unit REP
freshness source date ADD
citation URL
read
accept
agent
D_{train}
source-tagged
ledger T
traces
P1 units T08
P2 units T11
P3 freshness T08
arbiter · budget 2
candidate
base + patch
paired replay
strict gain?
D_{val}
fixed cases
D_{test}
sealed
next epoch
reject
Epoch loop
replay T
prev. vs current M
improved
regressed
unchanged
proposer context
rejected
blocks exact retry IDs
State
frozen weights
fixed during selection
updated online
(b) Edit arbitration
T08
source 12 ms · 2026-10-20
answer 12
unit and date omitted
T11 table column
latency (ms)
12
priority · evidence
P1 REPLACE units 4
value + unit T08
P2 REPLACE units 3
header only T11
P3 ADD freshness 2
source date T08
conflicts
P1 P2 P3
× same key
budget 2
P1 P3
(c) Replay on case V01
source 7.5 ms · 2026-10-18
v12
P2 only
v13
unit
7.5 ms
7.5
7.5
7.5 ms
date
2026-10-18
—
—
2026-10-18
citation
URL
URL
URL
URL
passed
1/3
1/3
3/3
baseline
1 = 1: reject
3 > 1: commit